

Homework: CIFAR-10 Image Classification From Scratch

Task Overview

In this assignment, you will build an image classification system for the **CIFAR-10 dataset** entirely **from scratch** using PyTorch. The goal is to help you understand the full deep learning workflow, including data preprocessing, model design, training, evaluation, and result interpretation.

Specifically, you will:

- Load and preprocess the CIFAR-10 dataset
- Design and implement your **own convolutional neural network (CNN) architecture**
- Train the model using supervised learning
- Evaluate the model's performance on unseen test data
- Visualize learning dynamics and prediction results

This assignment emphasizes **understanding and reasoning**, not copying existing architectures. Your design decisions should be intentional and justified.

Required Components

1. Data Pipeline

You must implement a complete data pipeline for CIFAR-10 using PyTorch.

This includes:

- Loading the CIFAR-10 dataset using `torchvision.datasets`
- Converting images to tensors and applying normalization
- Using `DataLoader` to create efficient mini-batch loaders with shuffling for training data
- Separating training and test datasets correctly

You are encouraged (but not required) to apply **data augmentation techniques**, such as random cropping or horizontal flipping, to improve generalization. If you use data augmentation, explain your choices briefly in the report.

Your data pipeline should be clean, reproducible, and clearly structured.

2. Model Architecture (From Scratch)

You must design and implement your **own CNN architecture**. You may not use pretrained models or copy architectures directly from tutorials.

Your model must include:

- At least **three convolutional layers**
- At least **one fully connected (dense) layer**
- Nonlinear activation functions (e.g., ReLU)
- Either pooling layers (e.g., max pooling) or strided convolutions for spatial downsampling

You are responsible for determining:

- The number of channels in each layer
- Kernel sizes and strides
- Where to place pooling layers

- The dimensionality of fully connected layers

The focus is not on achieving state-of-the-art accuracy, but on building a **reasonable and well-structured CNN** that demonstrates understanding of convolutional networks.

3. Training

You must implement a complete training loop for your model.

Your training setup must include:

- **Loss function:** CrossEntropyLoss
- **Optimizer:** Stochastic Gradient Descent (SGD) or Adam
- Training for **15–30 epochs**
- Proper switching between training and evaluation modes

During training, you must track:

- Training loss over epochs
- Training accuracy over epochs
- Validation or test accuracy over epochs

Your code should clearly show:

- Forward pass
- Loss computation
- Backpropagation
- Parameter updates

Ensure your training loop is efficient, readable, and logically organized.

4. Evaluation

After training, you must evaluate your model on the CIFAR-10 test set.

Your evaluation must include:

- Reporting the **final test accuracy**
- Plotting **training loss vs. epoch**
- Plotting **accuracy vs. epoch**
- Displaying a **confusion matrix** to analyze class-level performance
- Showing example images that are:
 - Correctly classified
 - Incorrectly classified

When visualizing results, focus on interpretation rather than aesthetics. Your plots should clearly convey learning trends and model behavior.

Deliverables

Submit one ZIP file containing:

- HW2_<YOUR_RIT_USERNAME>.ipynb or .py
- HW2_<YOUR_RIT_USERNAME>_report.pdf – short write-up summarize your process, analysis of your results, and sources you used (include ChatGPT and prompt if you used it)
- README.md

Grading Rubric (100 Points)

- Data pipeline correctness: 15 points
- CNN architecture (from scratch): 20 points
- Training loop correctness: 20 points
- Test accuracy $\geq 60\%$: 15 points
- Plots and visualizations: 15 points
- Report quality: 15 points

Bonus (up to +5 points):

- Test accuracy $\geq 70\%$
- Effective data augmentation
- Architecture ablation or analysis